NYC Homes × NYC Streets

Team Evaluation Guide

Housing Maintenance Code Complaints (H6, HPD) × Street Construction Permits (T3, DOT), New York City 2022–2025

Data Engineering and Visualization · Winter 2026 · Milestone 1
German International University · Faculty of Informatics and Computer Science

What this guide is for: every team member should be able to explain the whole project, defend every cleaning and integration decision, present their own two research questions, demo the website and answer the evaluators' questions. Read sections 1–3 together, then each member studies their own RQs (section 6) and the Q&A (section 9).

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1. The project in one page

Question. Are street disruption (DOT street construction permits) and housing distress (HPD housing maintenance problems) concentrated in the same NYC neighborhoods? If so, does street work come before indoor problems (a mechanism), or do both simply reflect the same kind of place (a shared cause)?

Answer. Same places, but no mechanism. Densities are strongly linked across 197 residential neighborhoods (Spearman ρ = 0.65). Every test of timing or change finds nothing: same-street gas work, 0–3 month lags, growth-on-growth, and out-of-sample prediction (R^2 0.924 → 0.925). The overlap is explained by shared neighborhood characteristics (dense, older residential housing stock), not by street work damaging homes.

Headline numbers to remember

Item	Value
H6 housing problems 2022–2025	3,160,386 downloaded → 3,153,693 after cleaning
T3 street permits 2022–2025	2,577,722 downloaded → 2,575,684 after cleaning (2,038 voided removed)
Permits placed in a neighborhood	2,508,047 in residential NTAs (97.4%); 42,132 in parks/airports etc.; 25,505 unplaceable
Integrated table	197 residential neighborhoods × 48 months = 9,456 rows, 119 columns
Spatial link	ρ = 0.65 city-wide; Bronx 0.82, Queens 0.77, Staten Island 0.65, Brooklyn 0.19, Manhattan −0.78
Double-burden neighborhoods	18 (12 expected by chance), West Bronx + Upper Manhattan
Seasons	complaints peak Dec–Jan (heat ×15 vs Aug); permits peak Mar and Jun
Prediction	own history + season R^2 = 0.924; + street-work lags 0.925; street-work only 0.13

The five parts of our story (the project description asks for exactly these)

- **The question:** see above. It matters because more than 600,000 permits a year open or occupy streets right outside buildings where tenants report 750k–850k problems a year.
- **The data:** H6 + T3 from NYC Open Data, connected by a spatial join (permit geometry → official 2020 neighborhood) and aggregation to neighborhood × month.
- **The discovery:** strong co-location, a reversed pattern inside Manhattan (commercial Midtown vs residential Harlem), opposite seasons, and no evidence of a street-to-home mechanism.
- **The limits:** correlation is not causation, permits are permissions and not observed work, monthly × neighborhood aggregation can hide very local or very short effects, complaints measure reporting, and there is no population denominator.
- **The impact:** a double-burden map and within-borough ranking for joint HPD/DOT outreach, and a finding that saves HPD effort: complaint forecasting needs only the neighborhood's own history and the heat season.

2. Requirements checklist (what the graders look for)

Every requirement from the Milestone 1 description, and where we meet it. Know where each one is so you can open it during the evaluation.

Requirement	Where it is	Status
One housing + one transport dataset, each ≥ 9 columns, with an integration attribute	H6 (33 cols) + T3 (39 cols); NB01 §4, automatic check §14	Done
Dataset overview: rows, columns, geo/time attributes, known issues, why chosen, how sizes are aligned	NB01 §4 + §4.1 table	Done
EDA with statistics and plots	NB01 §5 (H6) and §6 (T3): missing values, categories, time, geography, duplicates, outliers	Done
Missing values: justify drop vs impute	NB01 §7.8 (impute from same building), §8.10 (fill from same street segment), log	Done
Outliers (IQR, domain rules)	NB01 §7.10 (days to close, log-IQR) and §8.6–8.7 (durations, sizes)	Done
Inconsistent data, standardized formats	NB01 §7.2–7.6, §8.2–8.3 (text, labels, dates, ZIP/BBL/coords, typo fixes)	Done
Remove duplicates	NB01 §7.7, §8.5 (exact, repeated IDs; renewals are not duplicates)	Done
Integration with justified key and granularity	NB01 §9.1 comparison table; key = NTA × month	Done
Post-integration cleaning: new NaNs, redundant columns, type mismatches	NB01 §10.1–10.4	Done
Visualizations validating cleaning	NB01 §12	Done
Prompts used + dataset links	NB01 §16, NB02 last section, README §10	Done
2 RQs + 2 charts per member (6 → 12 + 12)	NB02, src/analysis.py, website tab "12 research questions"	Done
Website: multiple dropdowns	Borough, Year, H6 type, T3 kind, Neighborhood	Done
Website: search mode	Free text → sets the dropdowns (e.g. "Brooklyn 2022 new buildings")	Done
Website: central "Generate Report" button	Rebuilds KPIs + 7 charts	Done
Variety of interactive charts incl. a cross-domain one	Map, scatter, timeline, lag bars, ranking, heatmap, box	Done
Story section	Website tab "Our story"	Done
Deployed on a free platform	Vercel-ready slim copy in `vercel/`	**TO DO**: deploy + test
GitHub repo with full source + README (setup, deploy, contributions)	README.md written	**TO DO**: push to the team repo
Contribution documentation	README §7, NB01 header, website "Team & data" tab	**Check** the roles are accurate
Submission email to ICS503Data@gmail.com by 9 Nov 2026	IDs, repo link, deployment link, dataset links	**TO DO**

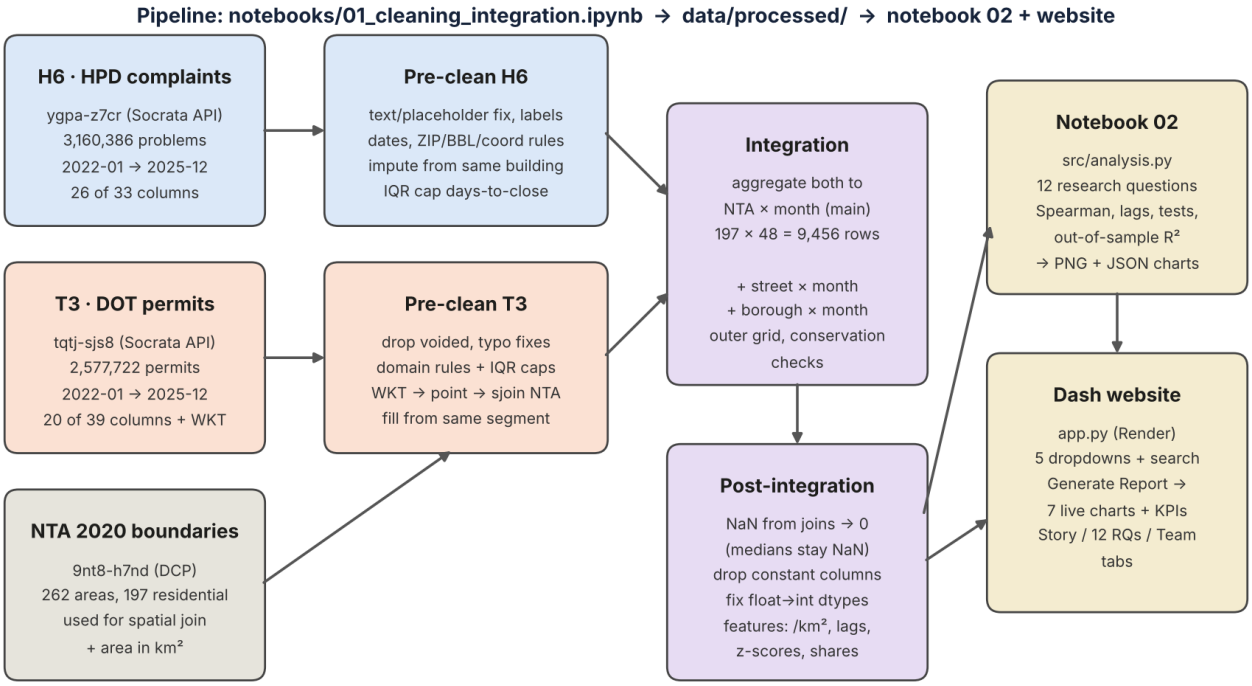
3. Datasets and why we chose them

	H6: Housing Maintenance Code Complaints and Problems	T3: Street Construction Permits (2022–Present)
Agency / ID	HPD · ygpa-z7cr	DOT · tqtj-sjs8
One row is	one problem inside a tenant complaint (a complaint can list several problems; mean 1.64)	one permit to open, occupy or build on a street or sidewalk
Size	~16.4M rows in total; 3,160,386 in 2022–2025; 33 columns	~3.99M rows in total; 2,577,722 issued 2022–2025; 39 columns
Time	received_date, problem_status_date	permitissuedate, issuedworkstartdate, issuedworkenddate
Geography	borough, ZIP, street, house no., lat/lon, BBL, **NTA name**	borough, on/from/to street, **WKT geometry** (State Plane feet)
Known issues	placeholders (N/A, UNKNWN), label variants (HEATING vs HEAT/HOT WATER), 6,363 negative close times, 15.9% HPD duplicate-report flags	~13% no geometry, typo COMMERICAL, 650 equipment spelling variants, impossible sizes (e.g. 102M sq ft), voided permits

Why this pair?

- Both are in the official catalog (H6 and T3). Both have time and geography, and both are large enough for neighborhood × month statistics.
- The connection is physical and plausible: gas, water and steam mains and building operations happen outside the very buildings tenants complain about. That gives testable hypotheses (heat ↔ gas work, plumbing ↔ water/sewer work), not just "count A vs count B".
- T3 only exists from 2022, so the window is 2022-01-01 to 2025-12-31: **4 complete years**, so every month and season appears equally often.
- **Why the full data and not a sample?** The description allows a filtered sample if justified. Our earlier starter sample took the first ~860 rows of each month, all from days 1–3, which makes every month identical in size and is useless for counts. The full download (paged, resumable, about 1 hour) removes any sampling bias. If asked: "we chose the full window because our questions are about counts and densities, and a sample would distort exactly those."

4. Architecture and the data pipeline, step by step



Step-by-step (notebook 01, section numbers in brackets)

Step	What we did and why
Download [2–3]	Socrata SoQL API with `$select`, `$where` (date window), `$order=:id` (stable paging), `$limit` 250,000, `$offset`. Each page is saved as parquet, so a crash resumes from the last page. Text columns are stored as category to save memory (H6 fits in 0.55 GB).
EDA [5–6]	Shape, dtypes, missing-value report, value counts, spelling-variant detection, monthly time pattern, coordinate/ZIP checks, duplicates, distribution of close times and permit sizes, geometry types.
H6 cleaning [7]	snake_case names; trim/UPPER/placeholders → NaN; semantic label merges (HEATING → HEAT/HOT WATER, VERMIN/PESTS → PESTS/VERMIN) and automatic spelling-variant merges; ISO dates; domain rules (5 boroughs, NYC ZIPs 100–104/110–116, coords inside NYC box, valid BBL/community board); duplicates; imputation **from the same building** (2,505 ZIPs, 264 coordinates/NTA names); NTA name → official code (100% matched; 99.93% agree with coordinates); 6,294 borough conflicts and 399 rows without neighborhood dropped; days-to-close: 6,359 impossible → NaN, log-IQR fence 180.9 days, 32,677 capped (1.04%).
T3 cleaning [8]	short readable names; typo COMMERICAL fixed; 650 equipment variants merged; 2,038 VOIDED permits removed (no work allowed); renewals kept (a renewal is a new legal permit); duplicates checked (0); domain rules (cuts > 1,000, sq ft > 5M, linear feet = 9,999 placeholder → NaN) then IQR capping (27,368 cut counts, 72 sq-ft values); work duration checked (0 inconsistent).
Geometry → neighborhood [8.8–8.10]	WKT text → shapely geometry → `point_on_surface` (always on the line) → EPSG:2263 feet; `geopandas.sjoin(predicate="within")` with the NTA polygons, plus nearest NTA within 300 ft for points on boundary streets. Permits with no geometry are filled from other permits on the **same segment** (on/from/to street, 260,709), same segment end (45,790) or same street (2,312), using only clear majorities (≥ 90%). 2,616 borough conflicts → UNMAPPED.
Integration [9]	complete grid of 197 residential NTAs × 48 months; H6 and T3 aggregated with 0/1 indicators + `groupby().agg()`; left-joined onto the grid. Conservation check: every problem and permit is counted exactly once.

Step	What we did and why
Post-integration [10]	NaN created by the join → 0 for counts (no complaint = zero complaints), medians stay NaN (a median of nothing is undefined, not 0); constant columns dropped (nta_type, is_residential); float → int for counts; key dtypes unified (datetime64[ns]); automatic asserts.
Features [11]	per-km² densities, shares and rates, log transforms, lags 1–3 and rolling 3-month sums, year-over-year change, within-borough-month z-scores, calendar features.
Outputs [13]	`integrated_nta_month.parquet/csv`, `street_month.parquet`, `borough_month.csv`, `nta_boundaries.geojson`, `data_dictionary.csv`, `cleaning_log.csv`.

5. Integration: key, granularity and post-integration cleaning

The two datasets share **no ID**. The catalog warns against forcing a row-level join "merely because a common attribute exists". We compared every option:

Option	Verdict
Row level (complaint ↔ permit)	No shared ID or address → meaningless; explicitly discouraged
Building / address	T3 house number is ~63% empty and T3 rows are street segments → too few matches
Street name × month	Useful for "same street" (RQ3), kept as an extra table; long streets cross many neighborhoods
ZIP code	T3 has no ZIP; ZIPs are postal routes, not neighborhoods
Borough	Only 5 units → too coarse (kept for totals and the website filter)
NTA 2020 neighborhood × month (chosen)	Official statistical units (DCP), stable boundaries, known area for per-km² rates; H6 has the NTA, T3 gets it by spatial join. Month smooths daily noise, shows seasons and allows lags

Why only residential neighborhoods?

65 of 262 NTAs are parks, cemeteries, airports and similar (ntatype ≠ 0). Almost nobody lives there (235 H6 problems) but they get 42,132 permits (e.g. airport roads). Keeping them would add rows with street work and no possible complaints, which biases every correlation toward zero. They stay in the borough totals.

Typical question: "Why left join on a grid and not an inner join?"

An inner join keeps only neighborhood-months that have both a complaint and a permit. Months with zero activity disappear, which is survivorship bias. The grid keeps all 9,456 combinations, and a missing count becomes a true 0.

6. The 12 research questions

Each member presents their two. For each: say the question, why it matters, the method in one sentence, what the chart shows, the answer with one number, and the limit. Everyone should know RQ1, RQ4, RQ5 and RQ12 because together they carry the main story.

RQ1 · Member 1 · Takiee-Allah (16001345)

Are neighborhoods with denser street construction also the neighborhoods with denser housing maintenance problems?

Why it matters. If the same places carry both burdens, city agencies could coordinate street work and housing inspections there.

Method. Totals per neighborhood for 2022-2025, divided by land area and years (per km2 per year). Spearman rank correlation across all residential neighborhoods, then again inside each borough to remove the borough-level confounder.

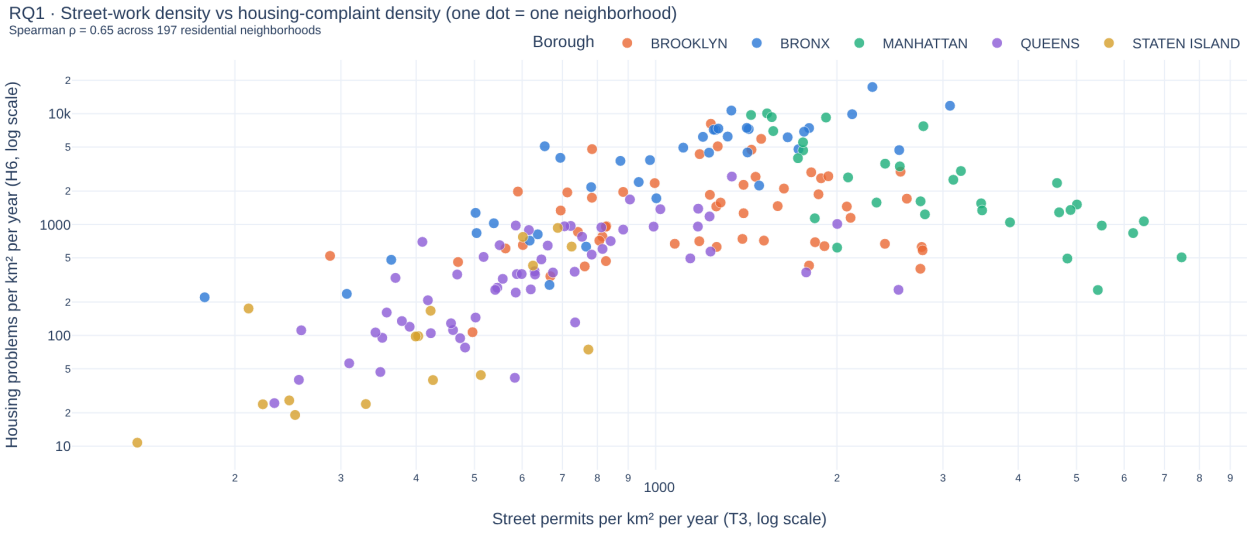


Chart choice. Log-log scatter (one dot per neighborhood, coloured by borough): both densities span several orders of magnitude, so log axes show the spread honestly.

Answer. Yes, strongly at city level: Spearman ρ = 0.65 across 197 residential neighborhoods ($p < 0.001$). The places with the most street permits per km² are also the places with the most housing problems per km². The within-borough picture is the real discovery, though. The link is strong in the Bronx (ρ = 0.82), Queens (0.77) and Staten Island (0.65), weak in Brooklyn (0.19), and **reversed in Manhattan (ρ = −0.78)**. Manhattan's most permit-dense areas are commercial cores like Midtown–Times Square (about 7,500 permits/km²/yr) with few tenants filing complaints (about 500 problems/km²/yr). Its complaint hot-spots, Harlem, Hamilton Heights and Washington Heights (about 9,000–10,000 problems/km²/yr), have only about 1,500 permits/km²/yr.

Limits. Both densities are driven by how built-up and residential a place is, which is a shared cause and not one causing the other. Density per km² ignores population and the number of housing units, because no population table was joined.

If the evaluator asks… If asked "why log axes?": densities span 10 to 17,000 problems/km²/yr; linear axes would squash 190 dots into a corner. If asked "why Spearman?": counts are skewed and the relationship is monotonic but not linear.

RQ2 · Member 3 · Ziad Khaled Saber Abdullah (17006438)

Do the two domains peak in the same months of the year, or in opposite seasons?

Why it matters. Timing matters for staffing: if streets are busiest when complaints are lowest, inspectors and crews compete less for the same months.

Method. City-wide monthly totals, each turned into an index (average month = 100) so both fit on one axis. Month-of-year averages identify each peak and low.

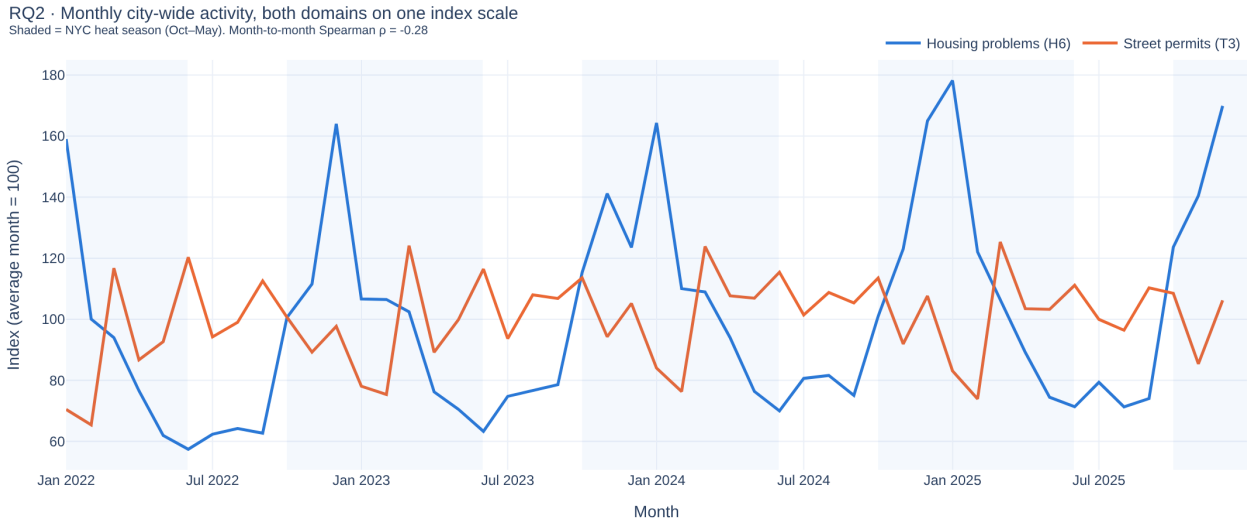


Chart choice. Two-line time series on a shared index scale with the legal heat season (Oct-May) shaded. A shared index avoids the misleading dual-axis effect.

Answer. They peak in different seasons. Housing problems peak in **December–January** (index about 155 and 152, average month = 100) and bottom out in **June** (66). The swing is driven by heat/hot-water complaints, which are about 15× higher in December than in August. Street permits are much flatter (73–123) and peak in **March** and June, with their low in **February**. Month to month, the two series are slightly negatively related (ρ = −0.28, p = 0.05). Street crews and housing inspectors are busiest at different times of the year.

Limits. This is city-wide, so it says nothing about any single neighborhood. Permits measure permission to work, not the days work actually happened.

If the evaluator asks… If asked "why not a dual axis?": two axes can be scaled to make any two lines look related. The index (average month = 100) puts both on one honest scale.

RQ3 · Member 2 · Kirolous Bassem (16002081)

On the SAME street, are heat and gas complaints higher in months when the street has gas-main work than in months when it does not?

Why it matters. Gas work can mean gas service shut-offs, which can take away heat, hot water and cooking gas from the buildings on that street.

Method. Street x month table (street names normalized identically in both datasets). Paired comparison: for every street that had months with AND without gas permits, compare its own average complaints in both kinds of months, separately for heat season and summer. Wilcoxon signed-rank test.

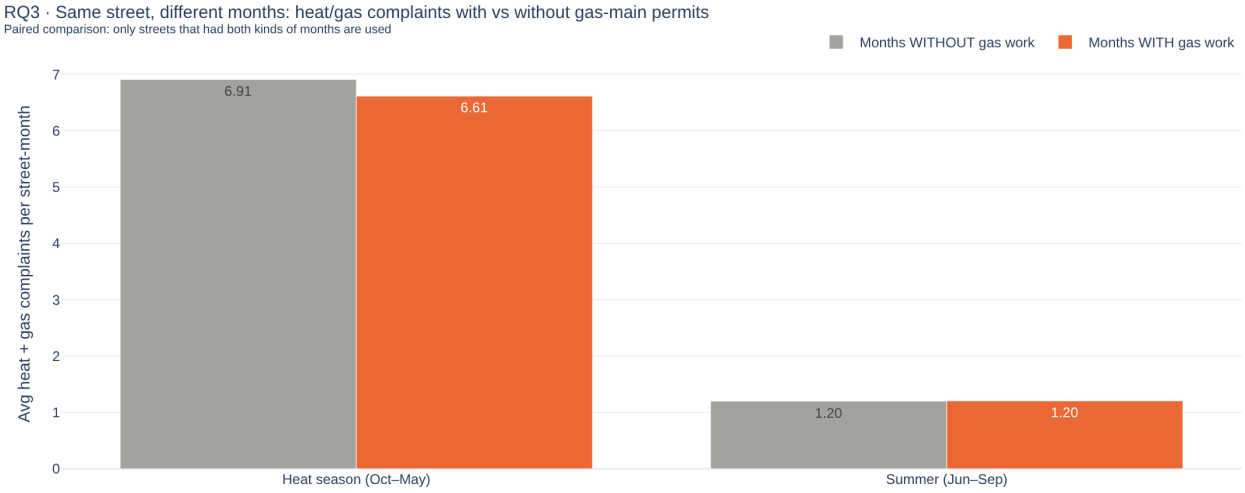


Chart choice. Grouped bar chart (with vs without gas work, per season): a direct visual of a paired before/after-style comparison.

Answer. No. In 4,505 streets compared with themselves during heat season, heat + gas complaints averaged 6.61 per street-month in months **with** gas-main permits and 6.91 in months **without**, slightly lower with work (ratio 0.96). In summer (3,962 streets) it was 1.20 vs 1.20, no difference. The simple story that gas work cuts off heat and causes complaints is not visible at street-month level.

Limits. Street names are matched by name, so a long avenue such as Broadway is one "street" across many neighborhoods. A permit does not say whether service was actually shut off. Very short outages may be fixed before tenants complain to HPD.

If the evaluator asks… Key word: **paired**. Each street is compared with itself, which removes street length, building count and income. The Wilcoxon test is the paired, non-parametric test.

RQ4 · Member 2 · Kirolous Bassem (16002081)

Does utility street work come BEFORE matching indoor problems (0-3 months later) in the same neighborhood?

Why it matters. A lead-lag pattern would let HPD anticipate complaint spikes from permits that DOT has already issued.

Method. For 4 work-to-problem pairs and lags 0-3 months: remove each neighborhood average and each month average (two-way demeaning), then Spearman correlation. Fisher 95% confidence intervals.

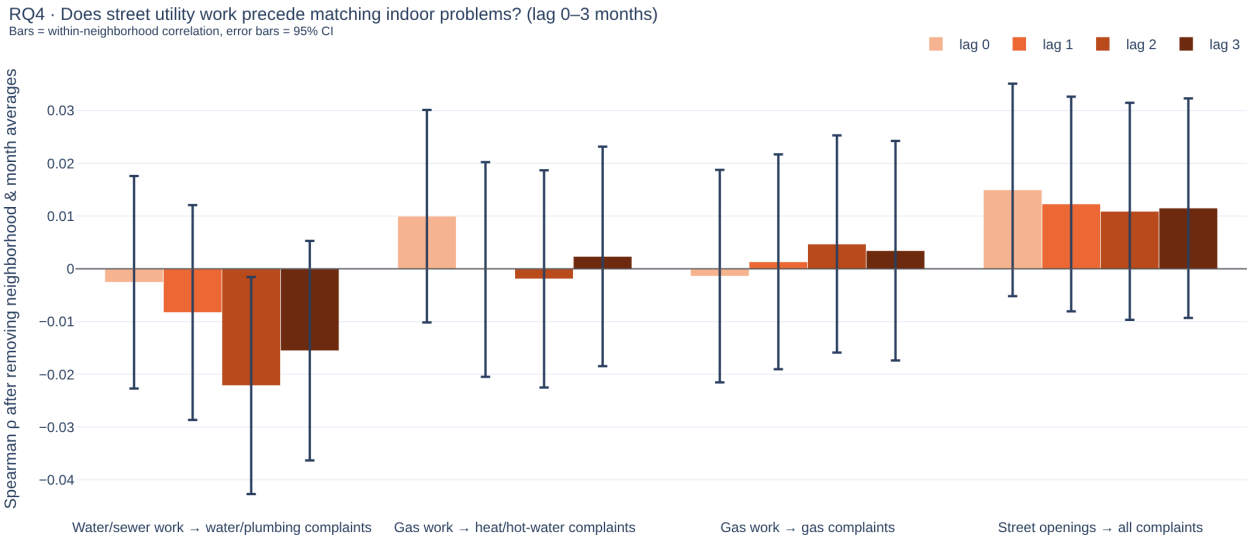


Chart choice. Grouped bar chart of correlation by lag with 95% error bars: shows size, direction and uncertainty together.

Answer. No timing link. After removing each neighborhood's own average and each month's city-wide average, every one of the 16 tested correlations (4 work→problem pairs × lags 0–3 months) lies between −0.022 and +0.015. All of them are practically zero, and 15 of 16 are not statistically significant. Unusual utility work in a neighborhood is not followed by unusual matching complaints there.

Limits. Monthly granularity could hide effects that last only days. Neighborhoods average many streets, so a very local effect could be diluted. A null result means "no detectable effect at this scale", not "proven zero".

If the evaluator asks… Key word: **two-way demeaning**. Without it, "busy neighborhood" and "winter" would create fake correlations. Live demo caveat: on a small website filter (e.g. Bronx 2024) the timing bar can show ρ ≈ 0.2. With only ~37 neighborhoods × 12 months that is within noise and multiple-testing range, while the full 4-year city test is ≈ 0.

RQ5 · Member 1 · Takiee-Allah (16001345)

Which neighborhoods carry a "double burden": top quartile for BOTH housing problems and street work per km2?

Why it matters. These are the places where residents live with both indoor housing problems and outdoor street disruption. That makes them candidates for coordinated action.

Method. Neighborhood densities for 2022-2025. Each neighborhood is classed by whether it is in the top 25% for H6, for T3, for both, or for neither. The number of "both" is compared with what independence would give (25% x 25%).

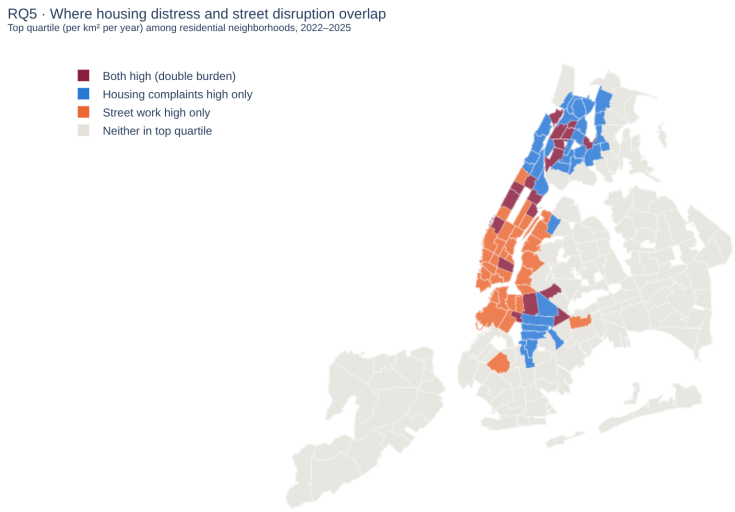


Chart choice. Choropleth map of the four classes: geography is the point of the question.

Answer. 18 neighborhoods carry a double burden (top quartile for both housing problems and street permits per km²). If the two were unrelated we would expect about 12, so overlap is about 1.5× chance. They are concentrated in the West/Central Bronx (Fordham Heights, Mount Hope, Mount Eden-Claremont, Concourse, University Heights, West Farms), Upper Manhattan (Harlem South, Inwood and others) and a few Brooklyn areas. On the map these form two connected clusters around the Harlem River.

Limits. Quartile cut-offs are a choice: a different threshold changes the count but not the clusters. Density is per area, not per resident.

If the evaluator asks… Why compare with 12.3? If the two rankings were independent, 25% × 25% = 6.25% of 197 neighborhoods would be in both top quartiles. We see 18, about 1.5× chance.

RQ6 · Member 3 · Ziad Khaled Saber Abdullah (17006438)

How many housing problems are reported per street permit, by borough and month of year?

Why it matters. The ratio shows where and when housing distress outweighs street activity, i.e. where the housing side of the story dominates.

Method. Borough x month totals (including permits that could not be placed in a neighborhood), pooled over 2022-2025, ratio H6 problems / T3 permits.

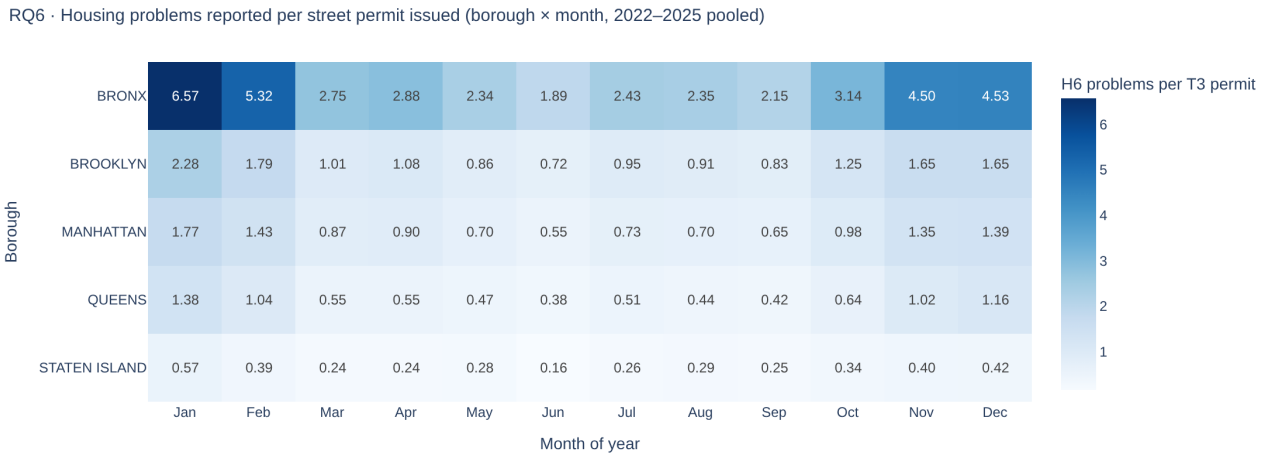


Chart choice. Annotated heatmap (borough x month): two categorical axes plus one value is exactly what a heatmap is for.

Answer. The ratio is highest in the **Bronx in winter**: 6.6 housing problems per street permit in January, against 1.9 in June. It is lowest on **Staten Island** (0.16–0.57). Every borough follows the same winter-high, summer-low pattern, because complaints swing with the heat season while permits stay flat (RQ2). The Bronx ratio is about 3× Brooklyn and about 3.5× Manhattan in every month. There, housing distress outweighs street activity far more than anywhere else.

Limits. The ratio mixes two different things (complaints and permissions), so it is a comparison index, not a rate. It is pooled over four years.

If the evaluator asks… The ratio is an index, not a rate (complaints and permits are different things). Its value is the comparison across cells.

RQ7 · Member 4 · Ahmed Hany (16004257)

Are housing complaints closed more slowly in neighborhood-months with heavy street work?

Why it matters. Street work may make buildings harder to reach or repairs harder to finish (e.g. waiting for utilities).

Method. Neighborhood-months with at least 5 problems. Street-work intensity = quartile of permit density compared with the same borough in the same month (z-score), so busy boroughs and seasons do not drive the tiers. Kruskal-Wallis test and Spearman correlation.

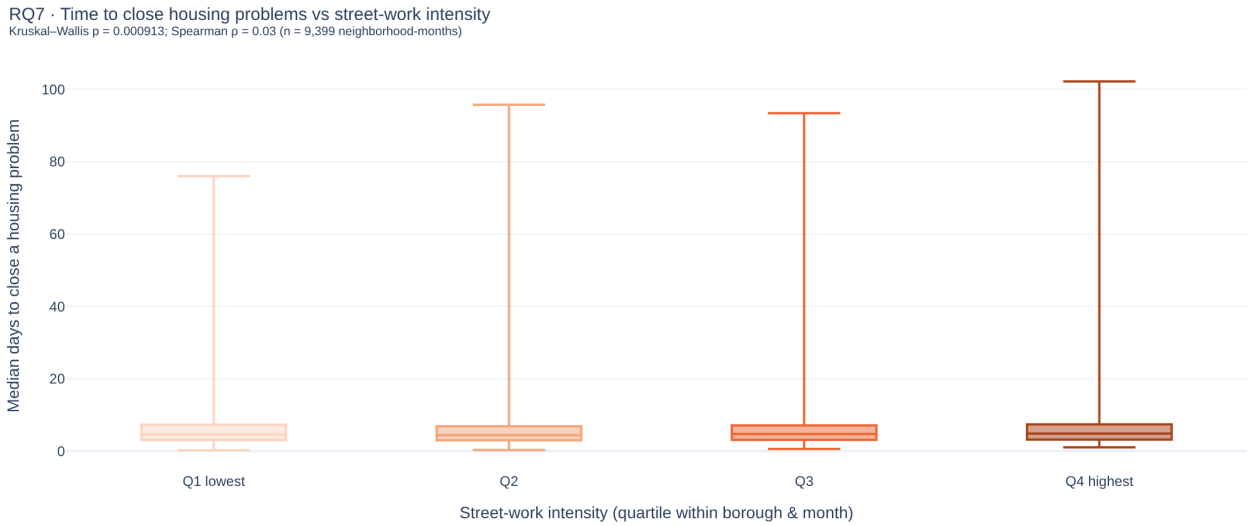


Chart choice. Box plot by intensity quartile: compares whole distributions of a skewed variable, not just means.

Answer. Practically no. The tiers differ statistically (Kruskal–Wallis p = 0.001, n = 9,399 neighborhood-months), but the size is tiny: the median time to close is 4.63 days in the lowest street-work quartile and 4.84 days in the highest. That is about 5 hours, and the correlation is ρ = 0.03. With 9,000+ observations even meaningless differences become "significant". Effect size is what matters here.

Limits. The value is the median per neighborhood-month, not per complaint. Closing a complaint depends mostly on landlords and HPD workload, which we do not observe.

If the evaluator asks… Key idea: **statistical vs practical significance**. With n = 9,399, a 5-hour difference is "significant" (p = 0.001) but meaningless. Always report effect size.

RQ8 · Member 4 · Ahmed Hany (16004257)

Which 15 neighborhoods are under the most combined pressure relative to their own borough?

Why it matters. A ranked shortlist is directly actionable for agencies with limited inspectors.

Method. Within-borough z-scores of log housing density and log permit density. The stress index is the average of the two.

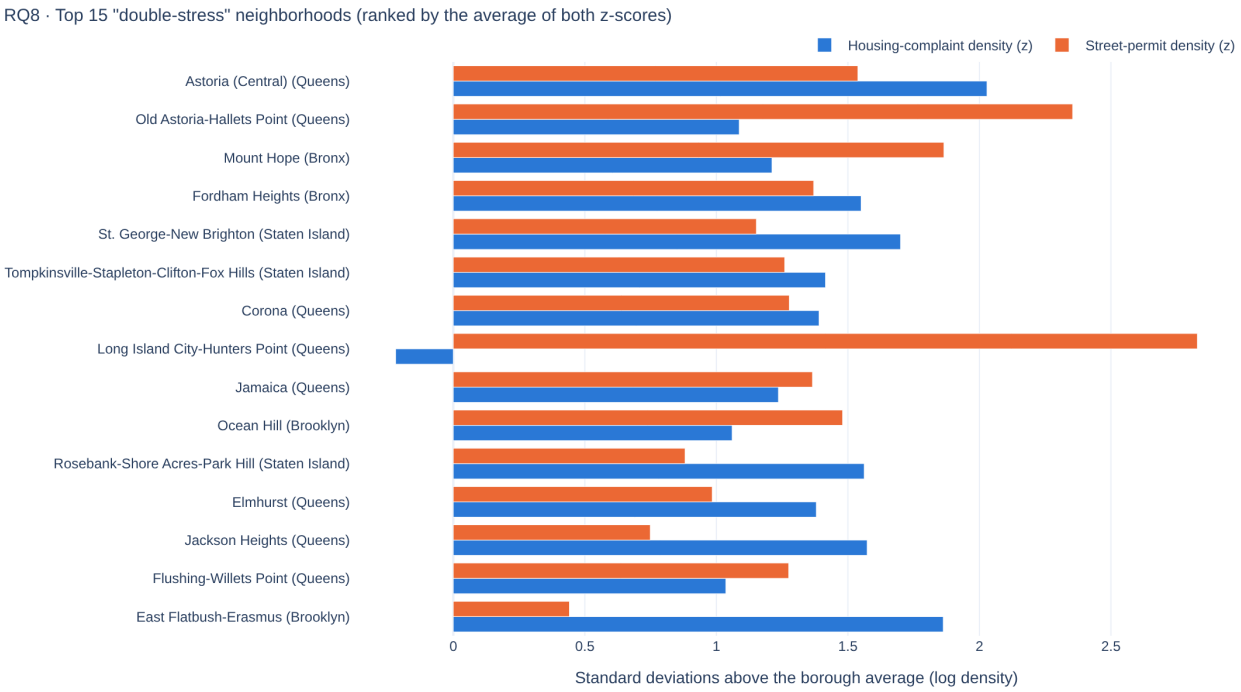


Chart choice. Horizontal grouped bar chart: long neighborhood names stay readable and both components stay visible.

Answer. Relative to their own borough, the most pressured neighborhoods are **Astoria (Central)** and **Old Astoria-Hallets Point** in Queens, **Mount Hope** and **Fordham Heights** in the Bronx, and **St. George-New Brighton** and **Tompkinsville-Stapleton** on Staten Island, followed by **Corona** (Queens). Using within-borough z-scores surfaces places a city-wide ranking would hide: Staten Island's North Shore is unremarkable city-wide but stands out strongly inside its borough.

Limits. The index weighs both sides equally. Long Island City reaches the top 15 almost only through street work (housing z below 0), so always read both bars and not just the rank.

If the evaluator asks… Within-borough z-scores answer "unusual for its borough", which is what borough-level offices act on. Read both bars, because Long Island City is high almost only on streets.

RQ9 · Member 5 · Abdallah Ahmed Mohamed (16004141)

Is the MIX of street work different in the neighborhoods with the most heat/hot-water complaints?

Why it matters. If high-heat-complaint areas have more gas or steam work, utility infrastructure could be a shared underlying factor.

Method. Neighborhoods split into 5 equal groups (quintiles) by heat complaints per km2. For each group: share of permits by work kind.

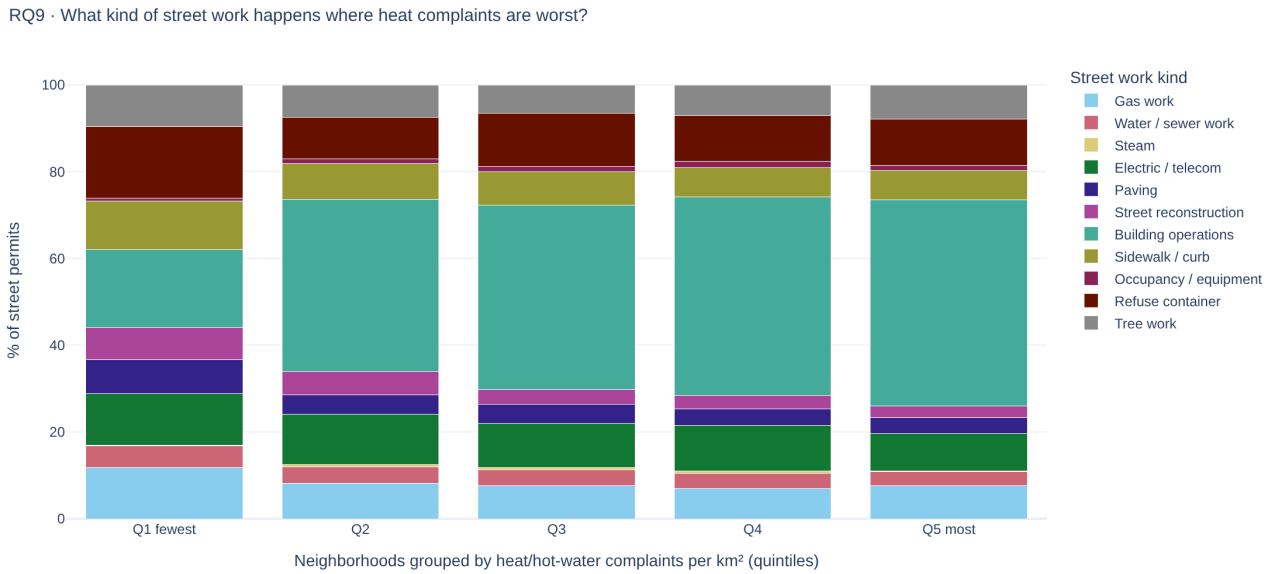


Chart choice. 100% stacked bars: compares composition across groups.

Answer. Yes. The mix shifts from utility work to **building operations**. In the 20% of neighborhoods with the fewest heat complaints per km², building-operation permits are 18% of all street permits. In the 20% with the most, they are **48%**. Over the same range, gas work falls from 12% to 8%, water/sewer from 5% to 3%, paving from 8% to 4% and sidewalk work from 11% to 7%. High-heat-complaint areas are dense residential areas full of building-side work (scaffolds, sheds, renovations), not areas with unusually much gas or steam work.

Limits. Work kinds are our grouping of DOT permit types (documented in notebook 01). Shares hide volume: dense neighborhoods have more of every kind in absolute terms.

If the evaluator asks… "Building operations" = permits for work next to buildings (sheds, scaffolds, construction). High-complaint areas are dense residential blocks, so more building-side work and proportionally less utility work.

RQ10 · Member 5 · Abdallah Ahmed Mohamed (16004141)

Did neighborhoods where street work grew the most (first vs last year) also see housing complaints grow?

Why it matters. Change-on-change removes everything that is constant about a neighborhood (size, building age, density). That is a stronger test than comparing levels.

Method. Percent change from the first to the last full year for both domains, for neighborhoods with a stable baseline (at least 50 problems and 20 permits in the first year). Spearman correlation and a quadrant count.



Chart choice. Scatter with quadrant lines at 0%: each quadrant has a plain meaning (both grew, both fell, ...).

Answer. No. Across 196 neighborhoods, the change in street permits from 2022 to 2025 is unrelated to the change in housing problems ($\rho = -0.01$, $p = 0.91$). Housing problems grew city-wide (+17%, from 732k to 854k), so most neighborhoods sit above zero whatever happened to their street work. Of the 108 neighborhoods where permits grew, 78 saw complaints grow. Of the 88 where permits fell, 60 still saw complaints grow. That is almost the same share.

Limits. A change between two years is noisy for small neighborhoods. A 4-year window may be too short for slower effects such as new construction attracting new residents.

If the evaluator asks… Change-on-change cancels everything constant about a neighborhood (size, age, income). That is a stronger test of a causal link than levels. It shows no relation.

RQ11 · Member 6 · Mohamed Ahmed (16008868)

Which housing metrics move together with which street metrics across neighborhoods?

Why it matters. A full matrix shows the whole picture at once and guards against cherry-picking one pair.

Method. 12 neighborhood-level metrics (6 per domain). Spearman correlation matrix. The cross-domain block is outlined.

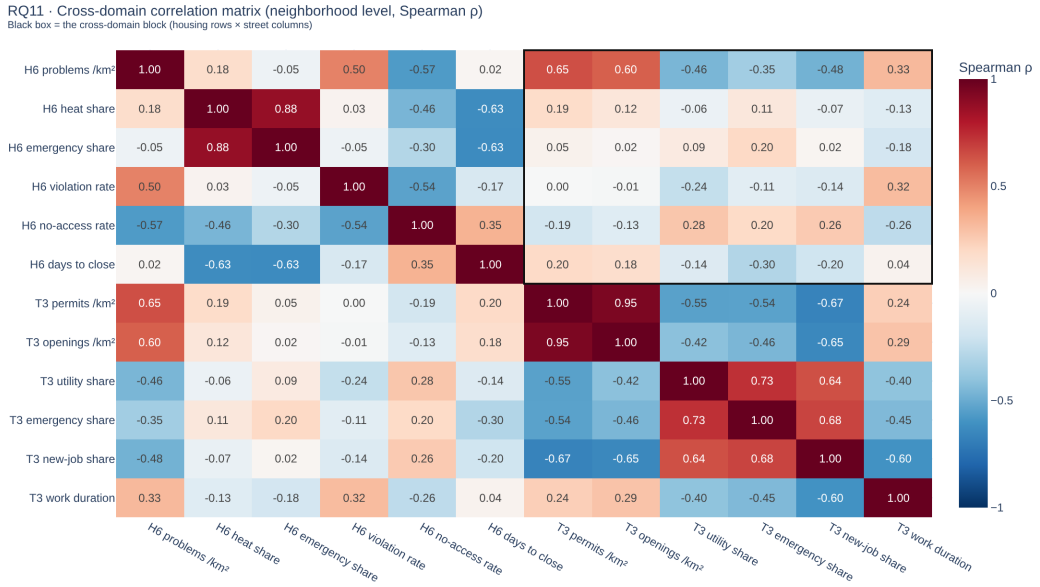


Chart choice. Annotated diverging heatmap (-1 to +1): the standard and clearest way to show a correlation matrix.

Answer. The strongest cross-domain link is simple density: housing problems/km² vs permits/km² (ρ = 0.65) and vs street openings/km² (0.60). Neighborhoods with dense housing complaints also have a lower share of utility (−0.46), emergency (−0.35) and new-job (−0.48) permits. Their street activity is more routine and more renewal-heavy. Inside H6, dense-complaint areas have higher violation rates (0.50) and fewer "no access" inspections (−0.57). Complaints there are more often confirmed, so they reflect real conditions and not just more reporting.

Limits. These are correlations between 197 neighborhoods (an ecological analysis). They describe places, not individual buildings or tenants.

If the evaluator asks… The black box is the cross-domain block. Within-domain blocks (e.g. heat share ↔ emergency share 0.88) show the metrics behave sensibly, which is a sanity check.

RQ12 · Member 6 · Mohamed Ahmed (16008868)

Does knowing the last 3 months of street work help PREDICT next month's housing problems in a neighborhood?

Why it matters. This is the practical test of the whole project. If street permits carry real signal, HPD could use DOT data for planning.

Method. Linear models on log(1+problems): (a) month-of-year only, (b) the neighborhood's own past problems (lags 1, 2, 3, 12) + season, (c) model b + street-work lags, (d) street-work lags + season. Trained on 2022-2024, scored on 2025 (out-of-sample R2 and mean absolute error).

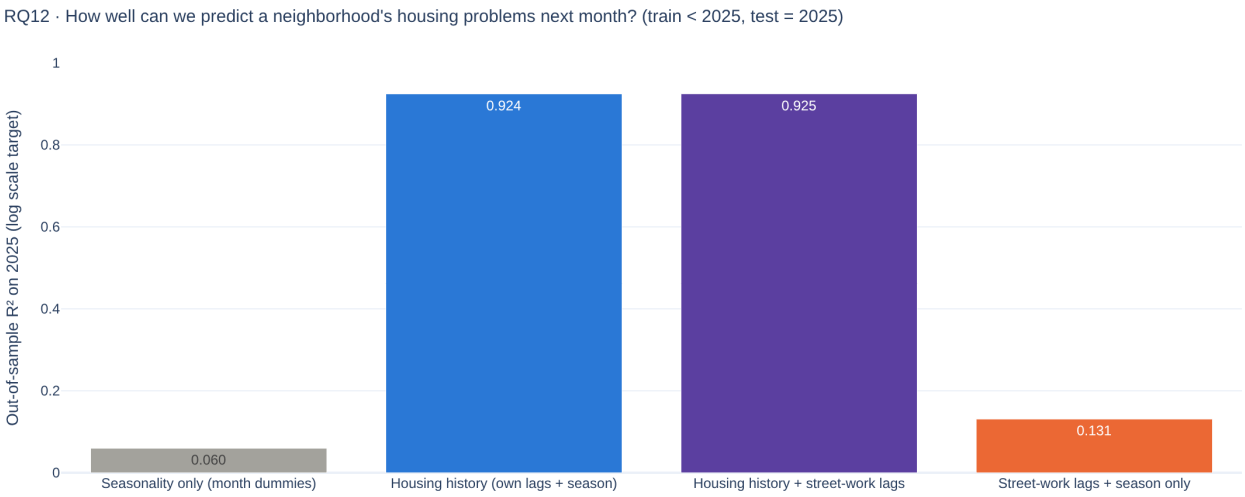


Chart choice. Bar chart of out-of-sample R2 per model: one number per model, easy to compare.

Answer. Street permits add no predictive power. A model using only a neighborhood's own complaint history plus season predicts 2025 out-of-sample with **$R^2 = 0.924$**. Adding 15 street-work lag features gives **0.925**, an improvement of 0.0001, and mean absolute error barely moves (67.6 → 67.5 problems). Street-work lags plus season alone reach only $R^2 = 0.13$. That is the same as "where is it dense", because permits also mark busy places. This is the clearest evidence that the spatial link in RQ1 comes from shared neighborhood characteristics, not a street-to-home mechanism.

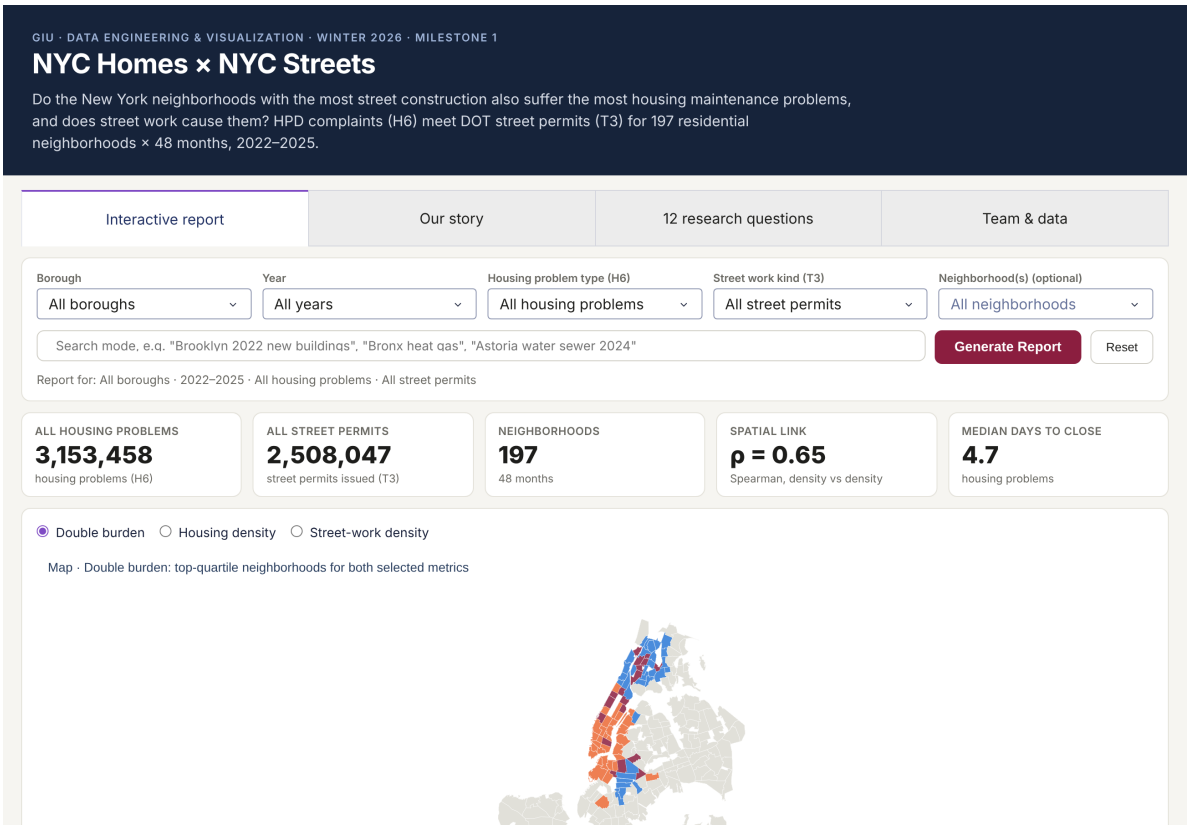
Limits. These are linear models on log counts. A non-linear model could find small effects, but the gap would have to be large to matter in practice. The test covers one year (2025).

If the evaluator asks… Out-of-sample (train 2022–24, test 2025) avoids overfitting claims. If T3 carried signal beyond "dense place", model 3 would beat model 2. It does not (+0.0001 R^2).

All 12 answers on one page

RQ	Member	Short answer
1	Takiee-Allah	Strong spatial link (ρ = 0.65), but reversed inside Manhattan (−0.78): commercial cores vs residential uptown
2	Ziad	Opposite seasons: complaints peak Dec–Jan (heat), permits peak Mar and Jun; ρ = −0.28
3	Kirolous	Same street with vs without gas work: no increase in heat/gas complaints (ratio 0.96 in heat season)
4	Kirolous	No lead–lag link at 0–3 months: all 16 within-neighborhood correlations ≈ 0
5	Takiee-Allah	18 double-burden neighborhoods vs 12 expected; clusters in West Bronx and Upper Manhattan
6	Ziad	Bronx in January has 6.6 problems per permit; Staten Island under 0.6 all year
7	Ahmed Hany	Close times differ by about 5 hours between street-work quartiles: significant but irrelevant
8	Ahmed Hany	Astoria, Mount Hope, Fordham Heights and the Staten Island North Shore top the within-borough index
9	Abdallah	High-heat-complaint areas: building-operation permits 48% of permits (vs 18%); less utility work
10	Abdallah	Growth in permits does not predict growth in complaints (ρ = −0.01)
11	Mohamed	Density is the only strong cross-domain link; dense-complaint areas have more confirmed violations
12	Mohamed	Adding street-work lags to prediction: R^2 0.924 → 0.925, no practical gain

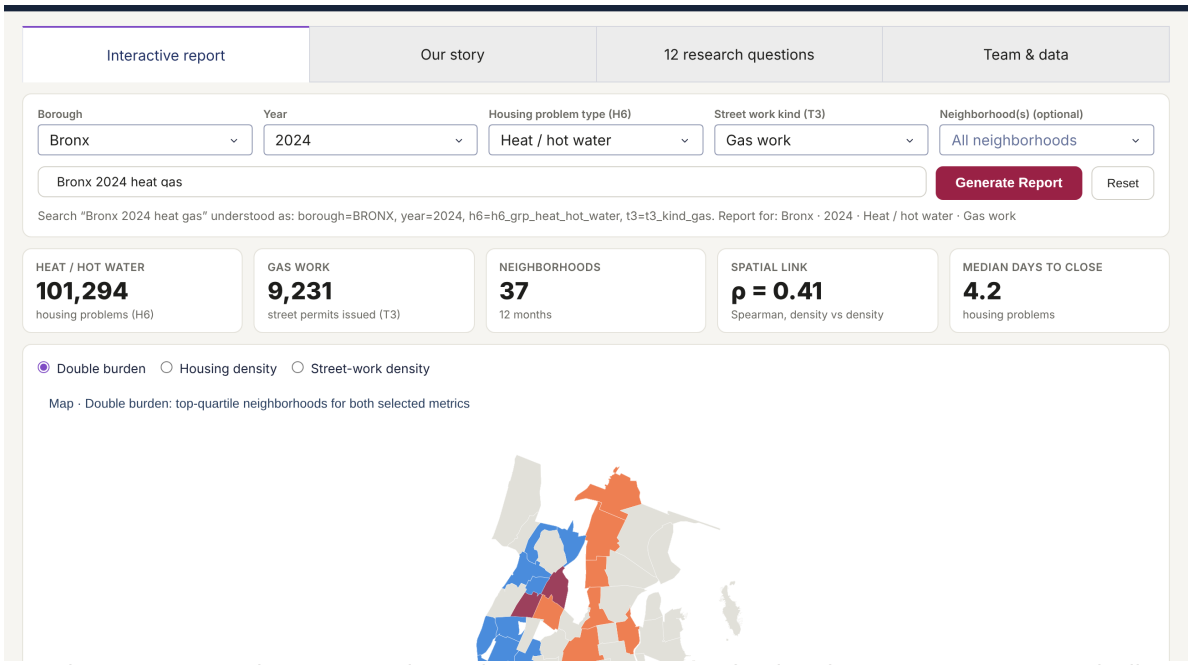
7. The website: features and a 3-minute demo script



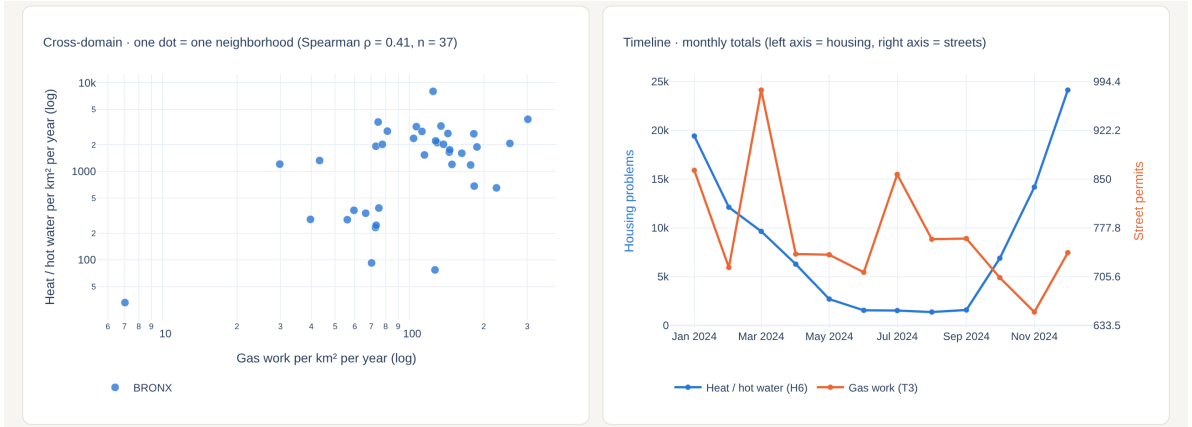
Opening view: header, tabs, five dropdowns, search box, the Generate Report button, KPI cards and the "double burden" map.

How it works (be ready to explain the code)

- `app.py` loads the processed tables once at start-up (`integrated_nta_month.parquet`, `borough_month.csv`, the NTA GeoJSON) and the 12 RQ charts saved by notebook 02 (`data/app/rq_figures/*.json`). **No data is simulated.**
- One central callback, `generate_report()`, is triggered by the **Generate Report** button, Enter in the search box, Reset, or the map-layer radio. It (1) parses the search text if present, (2) filters the table, (3) rebuilds the KPIs and all 7 figures.
- Search mode: `parse_search()` matches boroughs and aliases (BK, BX, "Staten"), a year, housing keywords (heat, leak, mice, mold…), street keywords (sewer, paving, scaffold → building, dig → street openings…), shared words (gas, water set both sides) and neighborhood names (Astoria → 4 NTAs). It then **writes the result back into the dropdowns** so the user sees what was applied, and shows a banner such as "Search … understood as: borough=BRONX, year=2024 …". A new search starts from the defaults.
- The H6/T3 category dropdowns choose which metric every chart uses (e.g. heat complaints vs gas permits). This is more meaningful than filtering rows, because the data is aggregated.
- Charts: choropleth map (double burden / housing density / street density layers), cross-domain log-log scatter with live ρ, timeline, timing test (lag correlations with 95% CI), top-12 ranking, borough × month ratio heatmap, box plot of days to close by street-work tier. All have hover, zoom and legend toggling.
- Tabs: Interactive report · Our story · 12 research questions · Team & data.



After typing "Bronx 2024 heat gas" and pressing Generate Report: the dropdowns were set automatically and the banner explains the parse.





Demo script (about 3 minutes)

- 1. Open the site. "This is NYC Homes × NYC Streets: 3.15M housing problems and 2.5M street permits, 197 neighborhoods, 48 months." Point at the KPIs: ρ = 0.65.
- 2. Map, "double burden" layer: "dark red = top quartile for both: the West Bronx and Upper Manhattan." Switch to housing density, then to street density: "Midtown is orange, not blue."

- 3. Type `Manhattan` → Generate Report → scatter: "inside Manhattan the link reverses ($\rho \approx -0.78$)."
- 4. Type `Bronx heat gas` → Generate Report: "this is our mechanism test. The city-wide 4-year result in RQ4 is ≈ 0." (See the RQ4 caveat about small subsets.)
- 5. Reset → Timeline and heatmap: "opposite seasons: complaints peak in winter, permits in spring and summer."
- 6. Tab "12 research questions" → scroll to RQ12: "street permits do not help predict complaints."
- 7. Tab "Our story": question, data, discovery, limits, impact. Finish with the impact sentence.

8. How to run, reproduce and deploy

Local run (Windows)

```
› python -m venv .venv
› .venv\Scripts\activate
› pip install -r requirements-notebooks.txt
› python app.py → http://127.0.0.1:8050
```

The processed outputs are in the repository, so the website runs right after cloning. To reproduce everything from the API:

```
› jupyter nbconvert --to notebook --execute notebooks/01_cleaning_integration.ipynb --inplace
  --ExecutePreprocessor.timeout=-1
› jupyter nbconvert --to notebook --execute notebooks/02_research_questions.ipynb --inplace
```

- Notebook 01 downloads about 5.7M rows (45–90 min, resumable). An optional free app token (`NYC_OPEN_DATA_APP_TOKEN`) speeds it up. `NYC_DATA_MODE=starter` runs an offline code check on the small starter CSVs (its counts are not real).
- Notebook 02 runs in about 1 minute and writes `visualizations/*.png` (needs `kaleido` for PNGs) and `data/app/`.

Deploy on Vercel (free)

- The `vercel/` folder is a slim copy of the website: same code and data, but without scipy and pyarrow so it fits Vercel's 500 MB Python limit. The live Spearman ρ uses pandas ranks (same value) and parquet is read with fastparquet.
- vercel.com → Add New → Project → import the GitHub repo → set **Root Directory** to `vercel` → Deploy.
- Test every control on the live URL. The first request after a quiet period is slower (cold start), so **open the link 2 minutes before the evaluation**.
- The map uses the free Carto basemap. If a network blocks it, set the environment variable `MAP_STYLE=white-bg`.

Repository layout

Path	Content
app.py, assets/style.css	Dash website
src/analysis.py	one function per research question (shared by notebook 02 and the website)
notebooks/01_cleaning_integration.ipynb	download, EDA, cleaning, spatial join, integration, post-cleaning, features (executed on the full data)
notebooks/02_research_questions.ipynb	12 RQs with charts, answers, limits, prompts (executed)
data/processed/	integrated tables, dictionary, cleaning log, NTA boundaries
data/app/	12 chart JSONs + story.json for the website
visualizations/	rq1.png ... rq12.png
tools/	text of the RQs/answers and the generators for notebook 02 and this PDF
docs/Team_Evaluation_Guide.pdf	this guide

9. Likely evaluation questions with model answers

Data & cleaning

Q: Why did you impute some values and drop others?

We impute only when another record gives the true value with near certainty: a building never moves, so its ZIP, coordinates and neighborhood can be copied from its other complaints (2,505 ZIPs, 264 coordinates). We drop only rows that cannot be placed in a neighborhood-month at all (399 H6 rows) or that contradict themselves (6,294 borough ≠ NTA borough). Other missing values (e.g. permit sizes) stay NaN because guessing them would invent measurements.

Q: How did you detect outliers?

Two layers. First, domain rules for impossible values: closed before received, a linear-feet value of 9,999 (a placeholder), more than 1,000 cuts, more than 5M sq ft. These become NaN. Second, the IQR rule (Q1 − 1.5·IQR, Q3 + 1.5·IQR) for extreme but possible values, applied on log(1+x) because durations and sizes are right-skewed. Those values are capped, not deleted, and flagged with an *_outlier column so nothing is hidden.

Q: Why cap instead of removing outliers?

The row is still a real complaint or permit and must be counted. Only its extreme measurement is limited, so one 1,400-day case cannot dominate a median or mean.

Q: What duplicates did you find?

No exact duplicates and no repeated IDs in either dataset (checked). H6 has an HPD "duplicate report" flag (15.9%): several tenants reporting the same problem. We keep those rows and flag them, because repeated reporting is itself a signal of severity. A unique-problems count is also provided. T3 renewals are not duplicates: they are new legal permits for continued work.

Q: What inconsistent data did you fix?

Old vs new HPD labels (HEATING = HEAT/HOT WATER), spelling variants (DOOR/DOORS, 650 equipment variants such as PORT-O-SAN), the typo COMMERICAL, placeholders like N/A and UNKNWN, leading spaces in street names, ZIP/borough mismatches (flagged), and invalid ZIPs, BBLs and community boards.

Q: Why remove voided permits?

VOIDED AFTER ISSUE means the permit was cancelled, so no work was allowed. Counting it would invent street work that never happened (2,038 permits).

Q: How did you handle the T3 geometry?

WKT text in NY State Plane feet → shapely geometry → a point that lies on the line (point_on_surface) → spatial join "within" the NTA polygons using an R-tree index, plus nearest NTA within 300 ft for points exactly on boundary streets. Missing geometry is filled from permits on the same street segment, only when ≥ 90% of those agree.

Integration

Q: Why NTA × month and not ZIP × year?

T3 has no ZIP, and ZIPs are postal routes. NTAs are statistical neighborhoods with known areas, so densities are possible. Months show seasons and allow 1–3 month lags, while years would leave only 4 points per neighborhood.

Q: What new problems did the join create, and how did you fix them?

Missing values: a neighborhood-month without permits gets NaN, which is a true 0 for counts, but medians stay NaN. Redundant columns: nta_type and is_residential became constant and were dropped. Type mismatches: counts became float after NaN and were converted back to int, and the key year_month was unified to datetime64[ns] before joining (a classic silent-join bug).

Q: How do you know the join did not lose or double-count rows?

Conservation asserts: the sum of h6_problems in the integrated table plus the excluded non-residential and unmapped rows equals the cleaned H6 row count. The same holds for T3. One row per key is asserted, and the grid size is asserted to be 197 × 48.

Q: What does the integration add that one dataset cannot?

Every RQ needs both sides: co-location (RQ1, RQ5), timing (RQ4), same-street pairing (RQ3), composition (RQ9), growth (RQ10) and prediction value (RQ12). It also produces a useful negative result: DOT data does not help forecast HPD complaints.

Analysis & honesty

Q: Your ρ = 0.65. Does street work cause complaints?

No, and we tested it. If it did, we would expect same-street effects, lags, growth-on-growth and predictive value. All four are ≈ 0. The correlation reflects shared characteristics: dense, older residential neighborhoods have more of both.

Q: Why per km² and not per person?

We did not integrate a population table (an honest limitation). Per km² still removes the biggest size effect. Per-resident or per-unit rates would be the next step, e.g. with census data.

Q: Why is Manhattan negative?

Its permit hot-spots are commercial and office districts (Midtown ~7,500 permits/km²/yr, ~500 problems) and its complaint hot-spots are residential uptown (Harlem ~10,000 problems/km²/yr, ~1,500 permits). Land use separates the two, which supports the shared-cause explanation.

Q: RQ7 is significant (p = 0.001). Why do you say "no"?

With 9,399 observations, tiny differences become significant. The effect is 4.63 vs 4.84 median days (about 5 hours) and ρ = 0.03. That has no practical importance.

Q: Did you check for multiple comparisons?

We report effect sizes and confidence intervals, not only p-values. In RQ4, 1 of 16 tests has $p < 0.05$, which is what chance alone would give ($16 \times 0.05 \approx 0.8$).

Q: Why Spearman rather than Pearson?

Counts and densities are highly skewed with a few giant values. Spearman uses ranks, measures monotonic association and is robust to outliers.

Q: Why is your prediction R^2 so high (0.92)?

A neighborhood's complaints are very stable from month to month and very seasonal. The point of RQ12 is the difference when street data is added (+0.0001), measured on a held-out year.

Q: Are your research questions "simple counts"?

No. They use densities, within-borough comparisons, paired same-street tests, lagged two-way-demeaned correlations, change-on-change, a correlation matrix and an out-of-sample model. The spec asks for "unexpected connections, spatial trends, or predictive patterns", and we have all three.

Website

Q: Show me the Generate Report button working.

Change Borough to Bronx and the H6 type to Heat/hot water, then press Generate Report. The KPIs, map, scatter, timeline, timing test, ranking, heatmap and box plot all update. Explain that it is one Dash callback with the button as the Input and the filters as State.

Q: How does the search work? Try "Brooklyn 2022 new buildings".

It sets Borough = Brooklyn, Year = 2022 and T3 kind = Building operations, and shows a banner saying what it understood. The parser uses keyword dictionaries with longest-match first ("hot water" beats "water") and neighborhood-name matching.

Q: Which chart shows the relationship between the two domains?

The scatter (H6 density vs T3 density with live ρ), the double-burden map, the timing test and the heatmap ratio.

Q: Is the website data real?

Yes. It loads the integrated table produced by notebook 01 from the full API download, and the 12 charts saved by notebook 02. (An earlier draft in the repo used generated demo data; it was replaced.)

Process

Q: Did you use AI?

Yes, Claude. The prompts are listed in notebook 01 §16 and notebook 02. Every output was run and checked by the team, and every number in the story comes from executed notebook cells.

Q: What would you do next?

Add population (ACS) to get per-resident rates; use daily data around individual gas-main permits within a radius of buildings (a sharper mechanism test); add a third dataset such as building age (PLUTO) to test the shared-cause explanation directly.

Q: What was the hardest part?

Placing 2.5M permits in neighborhoods: 13% had no geometry, and points on boundary streets fall between polygons. The segment-based filling and the 300 ft nearest rule solved it, with 97.4% placed.

10. Statistics cheat-sheet

Concept	Plain explanation	Used in
Spearman ρ	Correlation of ranks, from −1 to +1. Measures "when one goes up, does the other go up?" without assuming a straight line. Robust to outliers.	RQ1, 2, 4, 7, 10, 11
Density per km² per year	count ÷ area ÷ years. Removes the "bigger place, more of everything" effect.	RQ1, 5, 8, 9, 11
Within-borough z-score	(value − borough mean) ÷ borough std of log density: "how unusual for its borough".	RQ7, RQ8
Two-way demeaning	x − neighborhood mean − month mean + overall mean. Leaves only "unusual for this place in this month", so place size and season cannot fake a link.	RQ4, website timing test
Lag k	compare street work in month t with complaints in month t + k.	RQ4, RQ12
95% CI (Fisher z)	range that would contain the true correlation 95% of the time; if it includes 0, there is no clear effect.	RQ4
IQR rule	outlier if below Q1 − 1.5·IQR or above Q3 + 1.5·IQR (IQR = Q3 − Q1), here on log(1+x) for skewed data.	NB01 cleaning
Wilcoxon signed-rank	paired, non-parametric test: is the "with" value systematically different from the "without" value for the same street?	RQ3
Kruskal–Wallis	non-parametric test whether several groups (quartiles) have the same distribution.	RQ7
Statistical vs practical significance	p-value = "is it unlikely by chance?"; effect size = "is it big enough to matter?". Large n makes tiny effects significant.	RQ7
Out-of-sample R^2	share of variation explained on data the model never saw (2025). 1 = perfect, 0 = no better than the average.	RQ12
Ecological analysis	conclusions about places, not individuals (no "ecological fallacy").	RQ1, 11

11. Limitations to state honestly

- **Correlation ≠ causation.** The co-location has an unmeasured shared driver (building age, density, income).
- **Permits ≠ work.** A permit is permission for a window of time. It does not record the days of actual work or a service shut-off.
- **Aggregation.** Neighborhood × month can hide effects that are very local (one block) or very short (a day).
- **Reporting bias.** H6 counts problems tenants report. Under-reporting differs between communities.
- **No population denominator.** Rates are per km², not per resident or housing unit.
- **Geometry imputation.** About 12% of permits were placed from other permits on the same segment or street, and 1.0% (25,505) could not be placed.
- **Time window.** 4 years. Slower effects (new construction → new residents) would need longer.

12. Before submission: open tasks and per-member checklist

#	Task	Owner
1	Push the finished project (including the `vercel/` folder) to the team repo github.com/ZiadKhaled-wb/Data-Engineering-and-Visualization-Project.	Ziad
2	Deploy on Vercel (Root Directory `vercel`), test every control, and add the live URL to README line 8.	Ziad / Mohamed Ahmed
3	Check the contribution table (README §7, NB01 header, website Team tab in `tools/rq_meta.py`). It is graded; adjust to what each person really did.	Everyone
4	Each member re-runs or at least opens notebook 02 and reads their two RQs and the Q&A section.	Everyone
5	Send the email to ICS503Data@gmail.com before 9 Nov 2026: all 6 IDs, repo link, deployment link, the 3 dataset links.	Team leader
6	On evaluation day, open the Vercel link 2 minutes early (cold start).	Presenter

Per-member preparation

- **Takiee-Allah** (16001345): RQ1, RQ5. Data role: Data loading from the Socrata API (paged, resumable download), dataset overview, EDA of both datasets.
- **Kirolous Bassem** (16002081): RQ3, RQ4. Data role: H6 pre-integration cleaning: text standardization, category harmonization, duplicates, missing-value imputation from the same building.
- **Ziad Khaled Saber Abdullah** (17006438): RQ2, RQ6. Data role: T3 pre-integration cleaning: date/duration rules, IQR outlier capping of permit sizes, WKT geometry to points, spatial join to NTAs.
- **Ahmed Hany** (16004257): RQ7, RQ8. Data role: Integration (NTA x month grid, street x month and borough x month tables) and post-integration cleaning and validation.
- **Abdallah Ahmed Mohamed** (16004141): RQ9, RQ10. Data role: Dash website: dropdown filters, search mode parser, Generate Report callback, map and chart components.
- **Mohamed Ahmed** (16008868): RQ11, RQ12. Data role: Feature engineering (densities, lags, z-scores), statistical testing, GitHub repository, Vercel deployment, README.

Generated from the executed notebooks (full 2022–2025 data). Every number in this guide comes from notebook outputs.